

Black Hat USA 2012 - Are You My Type? Breaking .net Sandboxes Through Serialization

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Description

By: James Forshaw

In May, Microsoft issued a security update for .NET due to a number of serious issues I found. This release was the biggest update in the product's history, it aimed to correct a number of specific issues due to unsafe serialization usage as well as changing some of the core functionality to mitigate anything which could not be easily fixed without significant compatibility issues. This presentation will cover the process through which I identified these vulnerabilities and provide information on how they can be used to attack .NET applications, both locally and remotely, as well as demonstrating breaking out of the

partial trust sandboxes used in technologies such as ClickOnce and XAML Browser Applications.

Transcript

Retrieved with yt-dlp in a container, which is the only route YouTube still answers. Expect transcription errors in type and method names.

[Music] [Laughter] you're in a palace 3 for the breaking things track and this is the ru my type presentation by James porot hi there sorry about the uh short delay um computer problems so what am I here to talk about I'm going to talk about uh the research I did which ultimately resulted in the Microsoft bulletin ms235 and this was issues related to net binary serialization and the ability to use that to attack remote services or break out of partial trust sandboxes um it's also worth pointing out that actually um some of these issues have only been fixed um have only been mitigated in not actually been fixed so there's still some fundamental problems in net which are sort of hanging around which cannot actually be remediated and an easy way so the obligatory company slide I'm afraid uh before I get on to the real Tech um I work for a company called context information security outside um out of the UK we also have offices in Germany and Australia uh but we basically do three three main things research which obviously sort of thing is an example of uh we also do our Assurance stuff which is your standard pen testing and that kind of uh typical security work and we also do response so if you've been hacked we'll help come in and try and help you and uh try and work out how they got in and hopefully stop them from getting in again anyway so what is calization it's uh sort of a mechanism to transform a data structure of some form into which can be then stored and transmitted and then later recreated at another time or location um so being a quote who said that well I just said that because I couldn't find a better quote than that so why would we care what what is it about calization which might be a bad thing security wise well we only have to look at other Technologies to sort of demonstrate uh the potential risks involved in serialization mechanisms so you have almost the canonical example of serialization vulnerabilities in Java I'm sure most people here probably everybody here knows about sort of java serialization issues if you were here in 2009 you may have seen the presentation on attacking interoperability done by sort of the ibmx force team uh about Comm serialization issues and also PHP is another example there's an unserialized method you can call and if the untrusted data gets sent to that uh method potentially bad things can happen now do net being quite a big framework has a number of different uh serialization mechanisms sort of built into the framework uh first and foremost is is the I format of serializers so binary and soap and they they've been around since the very first version of net and also included in version one was the XO calization mechanisms and this is a bit more sort of binary ization is very very powerful and which is sort of why I'm talking about it xmo calization is considerably less powerful you can only create public types very very limited but in the scenarios in which you want to use it for it's quite quite useful and so there's some others uh as part of version three Windows communication Foundation introduced the data contractor serializers and 3.5 also had a built-in Json serializer but as I've said before I'm here to talk about B serialization now this is sort of baked in to The NET

Framework basically the sort of special support for it but not every object can be serialized if you for example were had a reference to say a file or some unmanaged resource it's not something you can just serialize so um at which point you actually have to say I actually want my C type or Visual Basic type to be serializable and to do this you must specify an annotation on that class which basically says hey I approve this for serialization and I will take responsibility for it so this is some very simple code which represents how you would actually do binary serialization and as you can see there's actually not much to it you create a a binary format instance and you then just push the object you want to serialize into it and out comes a binary blob of data so you can see why it's its power its flexibility is quite high and obviously that can lead to uh substantial misuse if uh if it's not taken care of now the end process of that as I say is a binary blob of data and this is sort of a rough representation of that class previously uh which we serialized so the binary data itself contains uh the name of a library so this is the dll effectively which hosted that type you then have a type name so this was the serializable class so that's recorded in there each field then is represented so each basically um variable um member variable you can specify in the class is then represented by a named value and then finally in this case the hello world string is stored as a as a straight binary blob in there but of course it's potentially it is a potentially dangerous operation the binary formatter can access private types private fields of an object and manipulate that and fill it could fill it with junk data so if you were to use binary formata in some sort of untrusted scenario so for example you were reading values you're reading B stream off a socket and pushing through binary form matter there's always a risk that there could be actually a security issue with that but surely the only time that's ever going to be problem is you get your object back and your badly written code does something stupid with that resulting object but it turns out that's not necessarily going to be the case so the framework potentially has mechanisms whereby code will execute during the process of deserialization and so in this particular case this bit of code which is doing derealization you've T taken potentially untrust input into the system and before you've even had an opportunity to stop that deserialization process happening you've already been done over and that's the sort of potentially risky scenario of of using this mechanism so one potential way that code can execute is by leveraging the custom serialization functionality sometimes you may not be able to specify exactly what you want to do just using the sort of default mechanism the default mechanism will just take each of your member variables and serialize those individually into the binary stream but sometimes that's not actually what you want so one of the mechanisms the framework provides is you can implement this I serializable interface and this interface allows you to specify custom serialization and deserialization code to run at that point of the process and in this case you've imp mented the get object data method which is run on serialization and you get a dictionary of uh key value pairs you can fill in with whatever data you like and then on the other side you implement a default a special Constructor which again takes that object and manually deserializes your object but you can see how potentially if there's dangerous code in that uh interface implementation something bad could happen just by the process of deriz ation so I did what anyone would do I wrote a program to basically hunt for bad stuff so in this case I wrote a tool first and foremost to go searching for serializable types and this is just an example of of five of the

main libraries in the framework and sort of the rough counts of the V of the number of classes which are serializable so for example mscore lib which is like the basic Library you have to have it in every net application is has about 681 serializable classes now considering there's only about 1,800 classes in the entire Library that's a fair proportion over a third of the classes is serializable now of those serializable classes 268 have custom serialization mechanisms which might mean some you could do something bad if you can find something bad in there now there's also other mechanisms for example callback so there's General sort of on deserialization run this bit of code do some checking that sort of stuff and there's also finalization now finalization is effectively destructors in the sort of C++ idea but for uh net and you'll see the similar if you know Java quite well you'll probably have encountered these as well and these are bits of code which will run to make sure you can clean up all your manage resources uh before the object gets disposed of in the garbage collector so in this case was two in mscore lib so let's actually go into finding some stuff which does something nasty if I just want to start and just be malicious well this class kind of fills in that um that requirement so you've got this class called temp file collection and it contains a list of uh file names which are the output of a compilation process so for example you can compile some sort of source code and it generates temporary files which it keeps a collection of and makes sure that they get deleted at some point in the future no guarantee when but it will definitely do it at some point in the future now this implements a finalizer and the thing the finalizer does is basically just walk through this list of files and deletes them so if an attacker can can serialize a list of files in there and hand it to you then when that object goes away it's going to start deleting arbit files on your file system even though you haven't actually interactive with the object at all so I'm just going to do a demonstration hopefully CH me this works uh possibly okay so this is on Windows 7 and I've got a serialized temp file collection class and right at the bottom this is our file we're going to delete important data. text on my desktop which is sitting there waiting to be uh be molested so if I run this bad application so this bad application just takes a file and pushes it through binary format it could have been a socket application or or any other sort of type of application which will take an arbitrary binary stream and if we load that file from the desktop it says we got a temp file collection object back that's great but it doesn't seem to have worked the file is still sitting there well that's because the finalizer isn't going to run immediately it waits until the garbage collector actually picks up and decides to destroy that object but if we actually close the application our file has now been deleted because it has to have run the finalizer to make sure everything is been cleaned up appropriately okay so there must be ways of protecting against this well you can be a bit more defensive in your in your development of any application which uses binary form matter and one of the ways is you can specify this special serialization binder class which allows you to basically just say I accept these types are valid and these types are not and the big one is just don't trust binary form matter with data coming from the outside world and you can also use plenty of other stuff as as I've shown with the previous slide there's a number of calization Technologies present in net there's things like protuff net which came out of Google so there's plenty of other other techniques you can use for serialization but of course you've done all this you've removed the uh use of binary formatter you've say used Proto buff instead so you're safe there's no

possible way you can be act well because it's such a fundamental uh serialization primitive it's also used in a few other ver Technologies which have been around since the very first version of net so if you're using net remoting for example you may be at risk or if you're using partial trust sandboxes you may be at risk and this is all happening inside the framework you don't even actually see it you don't have any reference to Binary formatter in your code but in fact you're actually using it or without um any actual knowledge so let's go a bit into how net remoting actually works because the first thing I'm going to look at is actually attacking remoting services so remoting Services has the concept of these application domains these are built into net they're fundamental net objects part of the the framework itself and act as isolation mechanisms for objects you cannot directly pass an object from one app domain to another without any other sort of mechanism in order to achieve that that feat now if you want to actually talk to a remote object so imagine one of these domains is actually sitting on a system somewhere the other side of the world for example connected over the Internet it will be publishing some sort of well-known service on a say a TCP port24 and another app domain wants to do some work on that so it creates first that TCP channel to act as the communication carrier and then it has to package up all the request data so it has to know who it's going to talk talk to at the other side and also all the parameters have to be packaged up as well so it packages that up sends it across the link and the well-known service will unwrap that and action that particular function call then obviously it's a symmetric process so you can pass it back any return value gets thrown back across the link and there is your remote call but that does still leave the question how parameters themselves get across the link because I've already said that an app domain you can't pass objects directly between app domains so in this case there's actually two ways in which you can pass objects which are built into the framework and if you've used any other sort of remoting technology such as RMI in Java you're probably very familiar with this sort of concept so the first is martial by reference so in this case your object never actually leaves your app domain but what it does do is it creates a special object which has all the information to refer back to the original value it is actually then that which gets converted to some binary blob and passed across the link where it can create a proxy object to talk back to the original object in your acon name but sometimes that's not appropriate so there's another way of do doing it you can if your object is serializable then it will actually get directly converted to a binary blob and when they gets ped across the link it actually recreates a copy of that object at the other side so any then function calls on that object will be local to that app domain but obviously I've talked about binary calization you can kind of guess what technology it uses to do this uh binary transformation so we had the malicious just the general malicious attack deleting files on your file system so let's go for something a bit more active um because it's a remote service we may want to actually try and do something particularly nasty to it so in this case I found a class called file info now file info as its name might suggest provides information about a file and it contains things like its path and its file size all that sort of stuff but during its custom deserialization process it first has to make sure that path is canonicalized so in to in order to get rid of dot do slash and all that sort of stuff it basically canonizes the path now it does this in a really really sort of simple way and just try try it doesn't touch the file system or it shouldn't touch the file system but this is actually a very

simplified version of that that code it basically splits the path into individual components and if one of those components starts with the tier character it assumes that perhaps that file name is actually a short path so a 8.3 window short short file name so in order to make sure it's got the definitely correct path it pushes it Through the Windows API of get long path name now get long path name has to refer to the original to the actual file system because otherwise how is it going to do that translation so what if we passed in a file name of SL eval SL tier share for example well what that ends up doing is your application starts making SMB requests back out and there's plenty of you can go to met exploit load up the smv relay module and you can then potentially exploit this so you can either reflect do credential reflection back to the uh person who generated that request or at least you used to be able to or you could relay that authentication mechanism to another system on a domain and attack something else that way but nothing is ever simple this entire process was anticipated by the original designers of The .NET Framework so this is a page in msdm and basically the first sentence is it's basically saying in order to actually work out whether the parameters you're passing are valid you must have deserialized these parameters and so you've got kind of a chicken egg problem and you could end up with automatic deserialization and as it quotes might try to exploit the moment of deserialization and you can you guaranteed that's exactly what I want to try and do so it implements type filtering so when I try and send my dodgy file info class across the link well the remote server just says no it doesn't match up with a set of rules it has in the framework which says these are valid objects to be serialized so we need a way of bypassing this and one of the things I found to do this is the data set class now the data set class is an in-memory database it can contain multiple tables but it has to do some clever manipulation of its serialization in order to handle the fact that it's able to serialize its tables both as XML and as binary blobs so what it does is it does serialization a two-step process so for example in this case we have a data set the star represents like our bad file info and the first thing the data set does is serialize that file info to a binary stream it then actually stores that in its outer wrapper serialized wrapper and it's that which gets passed across the link and when the deserialization process it um gets processed in the type filtering binary formatter in fact all it serializes is an array of bytes which is perfectly legitimate for um the secure lock down system but then the actual data set deserialization code kicks in it creates a privileged binary formatter and proceeds to de-serialize your file info class which then ends up you doing an SMB request out and potentially doing something nasty so I'm going to demo it it's got It's a demo on xps2 just because that's prior to the fix Microsoft uh on SMB reflection it just makes the demo look slightly nicer um but you can still use it for things like information gathering because it will go out and say who's actually run that piece of code or you can use it to sort of like Brute Force passwords if they're still running operating systems generating the landman hashes so uh I have here my xps2 machine expand that out slightly and on this machine I'm running a very very simple server I think there's the actual remoting code is about five lines something like that so there's no actual implicit sort of functionality being used which says I'm using binary formatter so if I uh if I actually run up a client for this so This is actually running under mono because mono implements basically um pretty much everything um according to the standards and so I can always I can just to demonstrate

it's actually working let me send just a simple message to the server and it says hello black cap on our so instead we want to um we want to run this SMB relay so I've got a copy of met sploit running running the SMB relay uh exploit payload so if we oops we go back to here and in this case we send our file info class so 108.1 so this is pointing back to the uh machine I'm hosting this on if we send a request now suddenly start seeing loads of stuff going on in the metp session and we get C to appear so fairly simple so again how can you protect against this well the main one or the certainly the Microsoft recommendation is just don't use net remoting net remoting is considered to be an obsolete technology and so Windows communication Foundation is the way to go also just don't expose it to the internet you can add authentication mechanisms to it as well to prevent just arbitrary people connecting to your service but it's also worth remembering that the process is symmetric so anything which works upwards towards the server probably also works down as well so if you can impersonate server in some way either booting DNS booting whatever you can then start attacking clients Instead at which point the authentication mechanism isn't necessarily going to help you so now on to the probably the most interesting bit the sort of breaking out of sandboxes now it's interesting that basically the the infrastructure which implements net remoting is done in exactly is used again for partial trust sandboxing that whole isolation aspect is a very useful feature to prevent people gaining access to objects outside the boundary of our app domain and you can also assign permission sets to these application domains which restrict the code what the code can do running in it so what will usually happen you'll have a host domain which will create this partial trust limited permission domain and that can then create untrusted classes coming from untrusted uh libraries somewhere and this is used in a few Technologies so for example there's click wants and there's also Zam or browser applications which are basically a browser hosted net program now there's various different permission sets you can specify so for example you can say the code can only access files in this directory or it can only open uh web requests to Google for example but there's also has this common with these sort of sandboxing mechanisms there's also some God level privileges the stuff which basically if you give it to untrusted code you may as well not have a sandbox at all so does things like can I call arbitrary Windows API functions if you can do that you can probably fairly easily break out the sandbox but unfortunately access to serialization Services is considered one of those God level privileges so there's no well unless you've implemented Ed very badly your sandboxing there's very little chance you'll actually have access to serialization in your partial trust code and that represents a problem but we know from remoting that all we actually have to do is find some way of getting an object to transition from one of those application domains to another at which point the framework will actually do all the work for us you'd think that would actually be quite a difficult thing to achieve but it's actually not as difficult as you'd imagine because the the boundary is fairly porous part of the use of net remoting is the fact that it's almost transparent you don't realize you're using it but that also then leads to the fact that you can't easily audit which bits are using umet remoting and which bits are not so for example in this case zamI browser applications had a bug with exception handling so the browser host would basically start creating a host application and in order to run the code in partial trust it would have to transition across that link into the other

code now this is still then the same flow of execution but our execution goes up and eventually leads to something we can control so in this case this is evil xap and someone's clicking a button but you don't actually have to have user interaction just running the code is sufficient if a piece of code creates an exception object um exception object are already serializable objects so it throws that and it starts going up the stack trying to find something to catch that exception to to stop the process from crashing and it goes all the way up to the the transition point between the app domains and the framework then goes nicely well no one in this app domain serviced that exception so maybe someone over the other side will do it and so it catches it throws it across the link and recreates it the other side and that gets us our roundtrip serialization and it is pretty much that simple this three lines of code is enough to get an arbitrary object roundtrip serialized in an Z browser application and this was one of the bugs which was fixed in ms235 but there's still a problem before we could just specify the binary stream itself so we could manipulate it in any way we liked manipulate internal states of objects to do any anything we wanted but we can't do that from partial trust code partial trust code is not supposed to be very privileged if you could manipulate the internal state of objects that's a bad thing so we cannot directly provide that binary stream but there must be a way of partial trust to manipulate that process and to do that we just have to go back to our I serializable interface which has been so fruitful already one of the things the serialization custom serialization process allows for is the ability to change your type now this might be used for example if you need to uh transport your object as a different type across a boundary and then when you deserialize you can then recreate your original type again so it allows you to just set your type it has a set type method but unfortunately there's no there was no actual restriction on what type you could specify so if I wanted to create a file info class I could just say set the type of that fill in some fake serialization data which would match up with what it was expecting and it would it would actually deserialize to that or uh deserialize into a different type to what it serialized from and this is just a sort of really simple demonstration so in our partial trust domain we have our exception and imagine our Green Dot is our custom serializable class it gets packaged up it's changed its type and actually now on the other side it's become a completely different type to what you put in now we could use this for various things we can obviously do it the SMB reflection stuff but let's face it that's not exactly the most interesting things to do especially as it's not something you can easily do now with the fix on reflection so I wanted a different way what if I could get hold of that object if I've just created a potentially corrupt and manipulated object if I can get it back and core methods on it because my part I got co-running in partial trust perhaps I can do something with that and so I found a class which just did that for me the evidence-based class was added in net 4 and is used to formalize the the mechanism of evidence I'm not I unfortunately don't have the time to actually describe what evidence is but it's basically sort of a use for trust decisions and it's marked as serializable but it also implements a method to do a deep clone of that object because it might need to copy these objects around a bit so this is the entire code for the Clone method and it looks fairly similar to code I've already shown up on the uh at the presentation so it's serializing the current object into a stream and then basically pushing that back through the binary format to spit out a new copy of that object but this wouldn't

have necessarily been a problem because if I called that because of the way net security works if the caller of that method didn't have permissions to do civilization it wouldn't actually make a difference but unfortunately it was annotated with a special override which basically says actually no no matter what people call me as I don't care what their permission was assume I have serialization permissions so that's a bit of a security problem so exploiting it is very simple if I derive a class from evidence base just create a new class fill it with a member variable which contains the object we want to get back but the custom serializable one so it will craft it if we called a clone method the roundtrip calibration process will ensure that in our own domain we've managed to get hold of that corrupt object and that's then we can do something with it so the fill in the final piece of the puzzle in terms of getting remote code execution the breaking out of a Sandbox we need a type which we can corrupt in such a way that it causes the security problem and for this I chose delegates now delegates are something which is sort of been built in since day one of the net runtime effectively are just sort of fancy uh function pointers but they have special um special treatment in the framework for performance reasons they effectively boil down to when they're jitted to straight function calls but it's also serializable which which makes it very very interesting especially as it also has an interesting property called multicasting now a delegate can Will normally just represent one call to one sort of method as sort of standard function pointer so in this case we've got a uh type of our uh delegate which is taking a um word-sized integer pointer but what we can do is we can actually join multiple Del together into one single delegate and then when we call that method both those functions gets called one after another with the same parameter but what if we could actually specify two different types of these delegates so in this case we now have a string delegate which takes a string but we also have our pointer one we try and combine those together and the framework says no because having those two types would cause type aling issues which basically could be exploited but what happens when we we do a serialized delegate what will this do if I actually call this method well it turns out it will do that so this is dump out of wind debug in this case ECX points to that fake value we specified because basically it's been passed on the stack there's been a type confusion issue and so effectively you've now just got a string which is represented by the pointer of 1 2 3 4 5 6 7 8 that ultimately leads to a vtable lookup and call and you can see from the stack that it's clearly massively confused as to what it's supposed to be doing it's calling through an in poter delegate into a string method and that shouldn't happen so I'm just going to do a demonstration of that in Click once so using the evidence-base class uh come on so we start this up make sure I have my web server running otherwise it's not going to do very much okay so if we run this from the browser we get various dialogues as it as it starts up but basically this is going to be uh creating a application on the desktop which is running with a heavily restricted permission set and you actually see this being used uh even now for example if you install Google Chrome via IE it will actually use this click WS mechanism to drop the installer on your system and install Chrome for you so obviously there's a there is a warning dialogue associated with that but we also get this big flag don't type your passwords into this application it's probably malicious we can see a big go button so this go button will effectively go and kick off the evidence-based vulnerability run some Shell Code by doing some corrupt delegates type aliasing all that

sort of stuff and we end up with [Applause] C but it doesn't really end there evidence base is a bug it's clear bug you shouldn't be able to either the method shouldn't have asserted the serialization permissions or you shouldn't have been able to derive your own object from it if you're coming from partial trust code and it was fixed it was fixed in ms235 so what if we could actually achieve the same result without any actual specific bug in in this sort of sense it would work in every of net CU obviously evidence base only came from inv version 4 and also crucially difficult to fix just because I'm feeling nasty so again we go hunting for interesting objects and one of the things I found is the hash table object or hashtable class and this is something which has been in since version one of the framework basically is a key value dictionary and was done before you actually had a proper generic dictionary which came along in net 2 and it serializes its keys and values but it doesn't actually serialize its hash table because due to the way in which it's implementing the hash function it cannot guarantee that those values will match up when you say deserialize from a file at a later date or on a different machine or a different version of the framework there's no actual guarantees in that to uh make sure that you're actually using the right thing so it calls a rebuild hashtable function and this thing uses a special object called an N quality compare or at least that's the interface which implements it and this can be specified by by code you can pass it in in the Constructor and say I want to use this special comparer as my mechanism to generate these hash values and so it gets serialized with the hash table because obviously once you get to the other side you actually want to uh use the uh the same thing you used at the other side and what it does is it goes through every single key in the collection and calls the get hash code method on this comparer which is interesting so what if this object was never actually serialized so this is how you can basically do it without ever having to find a specific book so in this case we've implemented our own custom equality comparer but crucially we've made it so it's a martial by reference object that means that object will never actually leave our partial trust sandbox so we throw our hash table across the across the boundary using the xbat uh bug I've already mentioned for example or any other mechanism to cross the boundary because there is generally there's still ways you can affect this and I get serialized across and recreated on the other side now obviously we've now got our bad object created in the other domain but obviously we can't access it because it's sitting in the host application domain but now the hashtable D calization code kicks in and this is where it goes interesting so it goes through each of its Keys passes it to the get hash code method but the get hash code method is implemented inside our martial by reference object so it has to serialize each of those keys back up and pushes them back across the boundary and because effectively we've broken the symmetry of the serialization process by doing this type aliasing issue when it goes back across the boundary it doesn't realize it's got to convert it back to our custom serialized object it now just thinks it's actually this corrupt delegate and you can use that to get back your object so final demonstration so in this case we've got a zaml browser application and you see it's hosted inside iie there used to be actually a Firefox Plugin or or MP API plugin for it although I think I believe that's now deprecated and come [Music] on taking it's time right so again just got a simple start button you can also specify the command if you're so inclined and to actually better demonstrate it in action I've also added a special marker object so this will basically pop

up a dialogue box saying at what stage of the process it's currently currently at so if they click Start we first get our object is being serialized in the partial trust ABD domain makes sense it's having to cross the boundary so we click okay it's now deserializing in the default app domain and this is the hosting app domain somewhere else on the system click again well now we're starting to go back the other way again now it's starting to serialize it again we finally get deserialization in the partial Trust app domain at which point we get out to appear and we can see from basically the fact that it's it's would now normally show this exception dialogue but as you can see it actually exploited it before it even finished the exception process because we were actually doing this exploit during the process the exception object was being deserialized so before the hosting code has even realized it's got an unquote exception to catch So It's tricky to protect against that because basically we've not technically used any real bug okay the exception trick is is a bug and has been fixed but there's other ways in which partial trust code can do that and it has massive potentials for back compat issues so effectively Microsoft fix was now you cannot do this type aliasing anymore if you are partial trust code you can no longer just arbitrarily set any any object type you like but also probably the biggest one is basically Microsoft have blocked any use of partial trust. net code from the browser because it was just too much of a security issue and so exam browser application are effectively banned certainly from the internet Zone uh you can still use click once but obviously you saw that dialogue which popped up and you could argue that if you click through a dialogue you could have clicked through a dialogue which says you want to get owned at which point it doesn't really matter so there was a lot more to the bulletin release than was actually just posted just in the cves so there was only two cves in it but there was actually quite a few different fixes implemented in that bulletin release now attacks from partial trust code have been mitigated but net remoting is still something which potentially you could attack if you found the right system and there's still a number of objects which if you could get them desized would do something potentially nasty so that's the end of the presentation uh if anyone's got any questions if you want more technical details the white paper should be on the CD which has code examples of how to create the the corrupt delegates and all that sort of stuff so there any questions no thanks very much